The LMS as a Build Target



The Problem

The Problem » Authoring in the LMS

- The LMS is a good place to **deliver** a course, a poor place to **author** one
- Authoring directly in the LMS makes it the *de facto* source of truth
- Consequences:
  - Content drift across pages and offerings
  - Limited, shallow version history
  - Format and platform lock-in
- Observed first-hand across several university offerings

CMS200 (N05127) Content Management Systems - Google Chrome

CMS200 (N05127) Conte

mylearn.torrens.edu.au/courses/22149

Trimester 1 HE, 2026

- Home
- Announcements
- Discussions
- Modules
- Online Classes & Chat
- Assignments
- Marks
- Reading List
- Library
- Collaborations
- People
- Attendance
- Course Analytics
- Lockdown Browser
- Quizzes
- Rubrics
- Files
- Outcomes
- Syllabus
- BigBlueButton
- Pages
- Settings

Account | Admin | Dashboard | Subjects | Calendar | Inbox | Study Assist | Careers | Help

Key Documents

Subject Outline

This document contains information about learning resources, assessments and subject learning outcomes.

CMS200_SubjectOutline.pdf

Minimise File Preview

Page 3 of 6 ZOOM

2.2 Subject Learning Outcomes

Subject Learning Outcomes	Mapping to Course Learning Outcomes
a) Develop and code dynamic webpages using web and database languages such as PHP and MySQL	Bachelor of UX and Web Design: CLO 3, 6, 8 Associate Degree of UX and Web Design: CLO 3, 6, 8
b) Develop and code modular web pages within a content management system (CMS)	Bachelor of UX and Web Design: CLO 3, 4, 8 Associate Degree of UX and Web Design: CLO 3, 4, 8
c) Analyse a client brief and user requirements to inform and impact website design	Bachelor of UX and Web Design: CLO 2, 5 Associate Degree of UX and Web Design: CLO 2, 5
d) Develop and code a custom website in response to a client brief using a CMS framework	Bachelor of UX and Web Design: CLO 3, 4, 8 Associate Degree of UX and Web Design: CLO 3, 4, 8

** The full list of Course Learning Outcomes and their descriptors is provided in Section 2.6 below.*

2.3 Assessment Tasks
12-week duration

Download (pdf)

Subject Planner

This document details what will be covered in each module and what activities will be used.

CMS200_SubjectPlanner.pdf

Minimise File Preview

a live Canvas subject: spot the retyped details ...

Assessment 1 - Submission - Google Chrome

Assessment 1 - Submissi

mylearn.torrens.edu.au/courses/22149/assignments/283483

Trimester 1 HE, 2026

- Home
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- Discussions
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- Online Classes & Chat
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- Settings

Account | Admin | Dashboard | Subjects | Calendar | Inbox | Study Assist | Careers | Help

Assessment 1 - Submission

Published | Assign To | Edit

Assessment Information

Download (pdf)

Minimise File Preview

Page 1 of 5 ZOOM

Subject Code and Title	CMS200 – Content Management Systems
Assessment	Dynamic Web Page
Individual/Group	Individual
Length	Web development folder; screen-recorded video
Learning Outcomes	The Subject Learning Outcomes demonstrated by the successful completion of the task below include: (a) Develop and code dynamic web pages using web and database languages such as PHP and MySQL (c) Analyse a client brief and user requirements to inform and impact website design
Submission	Due by 23h55 AEST/AEDT, Sunday at the end of Module 4
Weighting	30%
Total Marks	100 marks

Task Summary

Create a single web page that uses PHP for data retrieval and populating page content, and HTML and

Assessment_1_Brief.pdf

Please refer to the assessment brief attached above for details on how to complete this assessment.

Click on the link below to submit your assessment

Create a 30–60-second screencast video of your products page in which you scroll down the page from top to bottom and describe the features of your page and how it meets the client brief as you do so. Save the screencast video in MPG format.

Related Items

SpeedGrader

Download Submissions

1 out of 1 Submissions Marked

Trimester 1 HE, 2026
Home
Account
Announcements
Admin
Discussions
Dashboard
Modules
Online Classes & Chat
Subjects
Assignments
Marks
Calendar
Reading List
Library
Inbox
Collaborations
People
Study Assist
Attendance
Course Analytics
Careers
Lockdown Browser
Help
Quizzes
Rubrics
Files
Outcomes
Syllabus
BigBlueButton
Pages
Settings

These languages work in conjunction with each other and are indivisible. PHP enables you to connect to and manipulate databases. MySQL is the most popular database system used with PHP. Together, PHP and MySQL provide the functionality of dynamic data delivery. CSS is the styling of PHP syntax on a dynamic webpage. Just as you style static web elements using CSS, on a dynamic webpage, you can modify colours, size, typography, borders, spacing, composition and more.

By using CMSs with more control, ability and customisation, designers are able to develop and manage websites with more ease due to the sophisticated User Interface and User Experience of the CMSs.

To locally run WordPress (a CMS tool), you can install Docker, an open-source tool that is free and easy to use. Docker allows you to set up a full WordPress environment and develop your own themes for dynamic data delivery. You can access everything in a regular browser, just like a live website; however, it is all on your local computer. In the real world, final website files should be transferred from your computer to your client's server.

Subject Learning Outcomes

This Module will help you achieve the following outcomes:

1. a) Develop and code dynamic webpages using web and database languages such as PHP and MySQL
2. c) Analyse a client brief and user requirements to inform and impact website design

Time Management

Your workload expectation for each Module is 10 hours.

12-Week Delivery:

- The 10 hours per module (per week) comprises 3 hours per week of facilitated study and 7 hours per week of personal study.
- The 3 hours of facilitated study consists of attending class and responding to facilitator feedback.
- You are also to allocate 7 hours to personal study. This includes essential time spent pre-reading and viewing materials, making progress on your assessments and completing the learning activities.

Assessment Progression

Familiarise yourself with the three assessment briefs associated with this subject.

Assessment 1 is due at the end of Module 4 (Week 4). Ensure that you install Docker and WordPress successfully by the end of Module 1.

Complete Learning Activity 2 to help you improve the look and feel of your final product page for Assessment 1.

clearly acknowledge any external sources, including code, libraries, plugins, images, fonts, or other assets. If you adapt content from tutorials, forums or generative AI tools, you must note the source in comments or in your README, clearly stating what you used, where, and its extent and purpose in your project.

Academic Integrity Declaration

I declare that, except where I have referenced, the work I am submitting for this assessment task is my own work. I have read and am aware of Torrens University Australia Academic Integrity Policy and Procedure, viewable online at http://www.torrens.edu.au/policies-and-forms

I am aware that I need to keep a copy of all submitted material and its drafts, and I will do so accordingly.

Assessment_1_Brief.pdf

Please refer to the assessment brief attached above for details on how to complete this assessment.

Click on the link below to submit your assessment

Create a 30–60-second screencast video of your products page in which you scroll down the page from top to bottom and describe the features of your page and how it meets the client brief as you do so. Save the screencast video in MPG format.

Move the screencast video into your products listing web page folder containing all your web page files.

Compress the folder as a ZIP file and name the ZIP file **CMS200_A1**_*lastname_initial*.**zip**

For example, CMS200_A1_ghrol_d.zip

Submit the zip file task via the **Assessment** link on blackboard for this class. The Learning Facilitator will provide feedback via the Grade Centre in the LMS portal. Feedback can be viewed in My Grades.

Academic Integrity Declaration

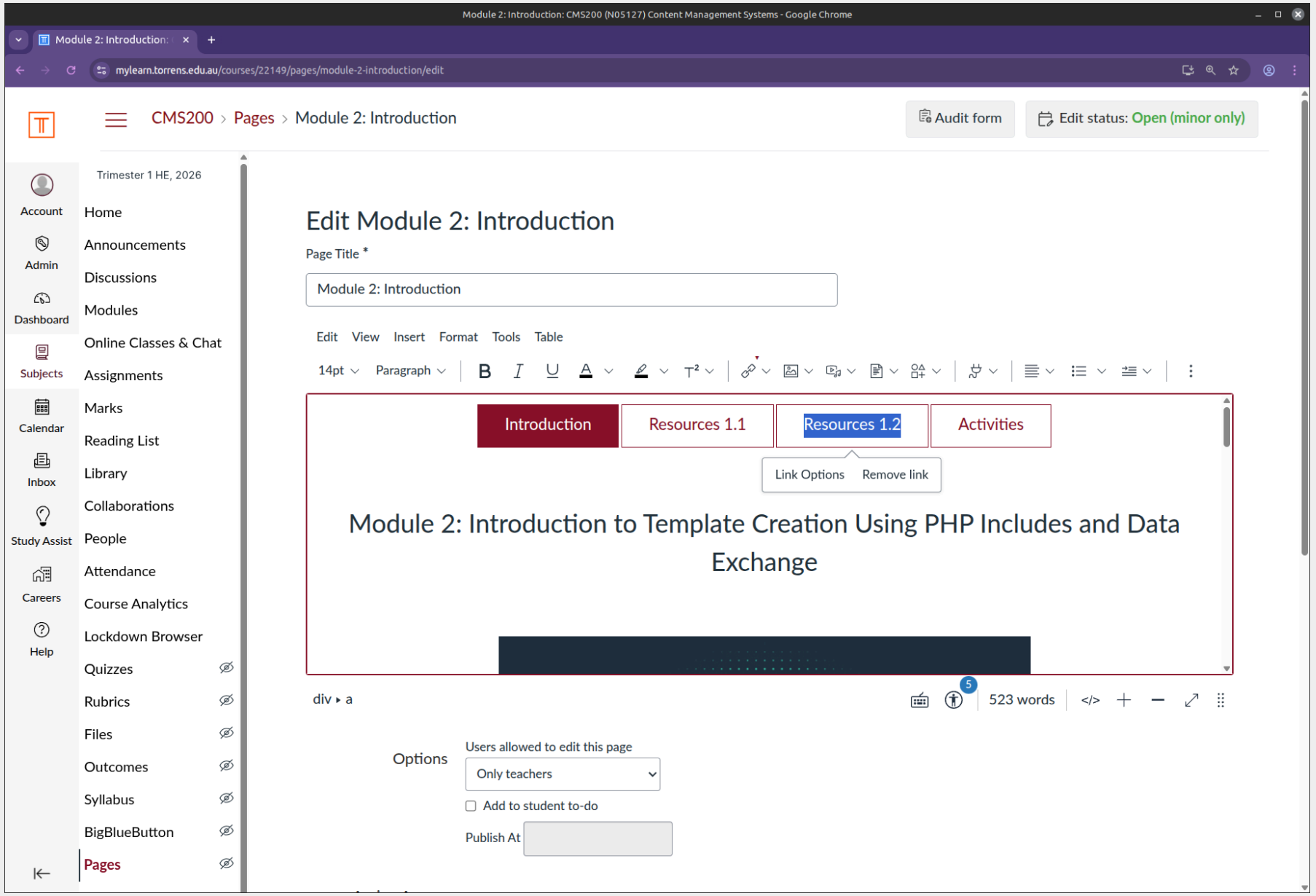
Academic Integrity Declaration I/we declare that except where I/we have referenced, the work I am/we are submitting in this attachment is my/our own work. I/we have read and am/are aware of Torrens University Australia Academic Integrity Policy and Procedure viewable online at http://www.torrens.edu.au/policies-and-forms. I am/we are aware that I/we need to keep a copy of all submitted material and their drafts, and I/we will do so accordingly.

The Problem » The Manual Chain

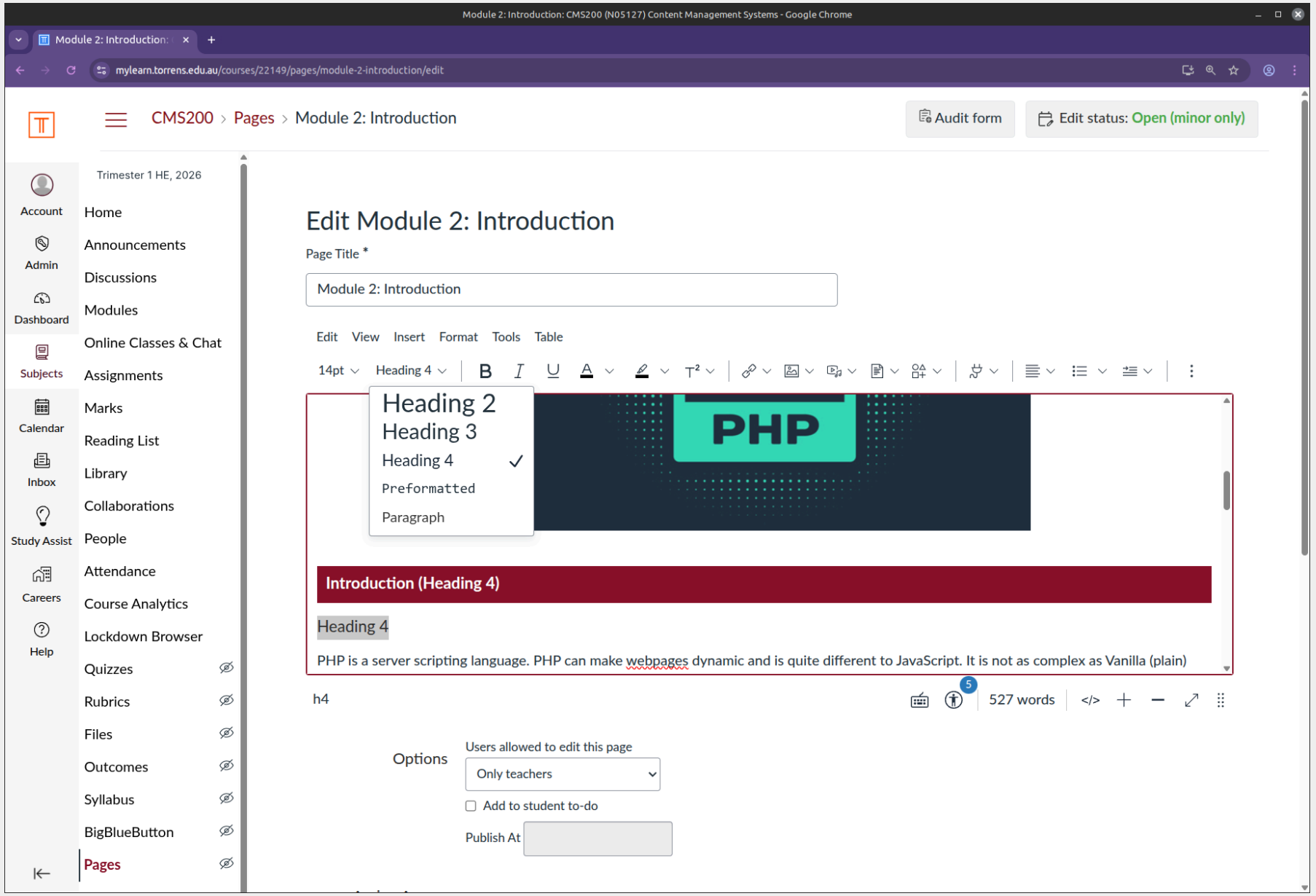
word documents → manual citation formatting → export to pdf
→ build lms structure by hand → copy-paste page by page
→ re-format in the lms editor → upload & embed assets
→ retype metadata in multiple places → cross-check → qa pass
→ rework after feedback → hotfix pages during delivery → drift

---

structured content → automated build → lms-ready output



the Canvas rich content editor: built by hand, page by page



hand-formatting, heading by heading, in the LMS editor

The Reframing

The Reframing » A Requirements List

Complaints reframed as priorities, before building anything …

- Currency of learning materials, ownership by domain experts
- Meaningful version history, single-sourced metadata
- Multi-format publishing, citation management
- Portability and accessibility

The Reframing » What If …

Instead of wrangling Word documents and repeatedly copy-pasting into the LMS editor, we:

write content once, in structured source files
then

The Reframing » What If ...

Instead of wrangling Word documents and repeatedly copy-pasting into the LMS editor, we:

write content once, in structured source files
then
automatically generate:

- 📘 LMS module pages
- 📄 assessment briefs
- 📑 formatted references
- ✨ and more

The Mental Model

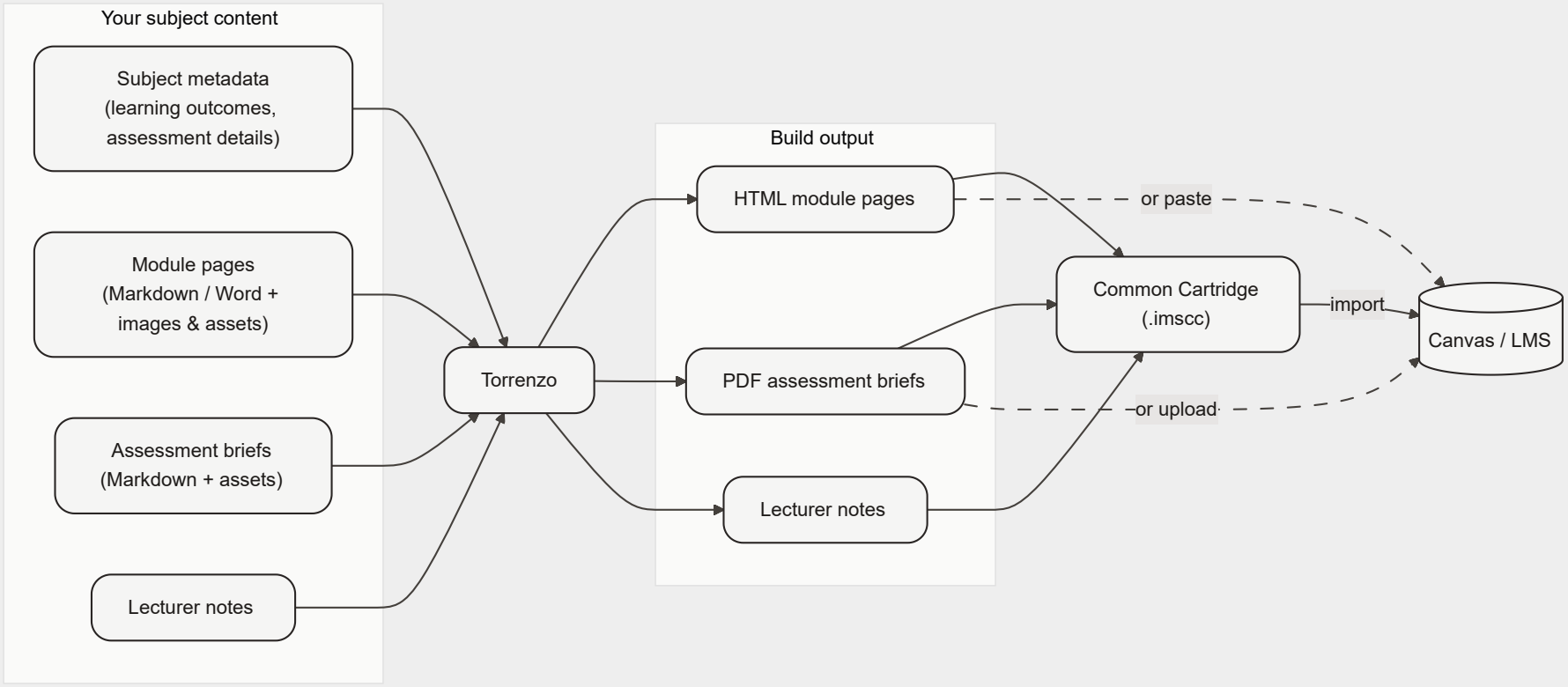
The Mental Model » Static-Site-Generator Thinking

- Same inversion documentarians use for docs portals:
  - The **repository is canonical**
  - A **build process** renders the required outputs
  - The **LMS becomes a deployment target**
- No bespoke platform -- just plain text and standard build tooling

The Mental Model » Standing on Familiar Ideas

- Publisher ecosystems: Manning, No Starch Press, O'Reilly
- Static site generators: Eleventy, Hugo, Jekyll, Pelican
- Docs-as-code: Docusaurus, GitBook, MkDocs, Sphinx
- Document transformation: LaTeX, Marp, Pandoc
- Build & publishing pipelines: GitHub Actions, Netlify

The Pipeline



sources in, build in the middle, LMS-ready artefacts out

The Pipeline » End to End

- Sources: **YAML** (metadata) + **Markdown** (content) + **BibTeX** (references)

Input	Output
Module pages	HTML module pages
Assessment briefs	PDF briefs
Teaching notes	HTML notes
Everything above	Common Cartridge (`.imscc`)

- The cartridge bulk-imports into **Canvas**

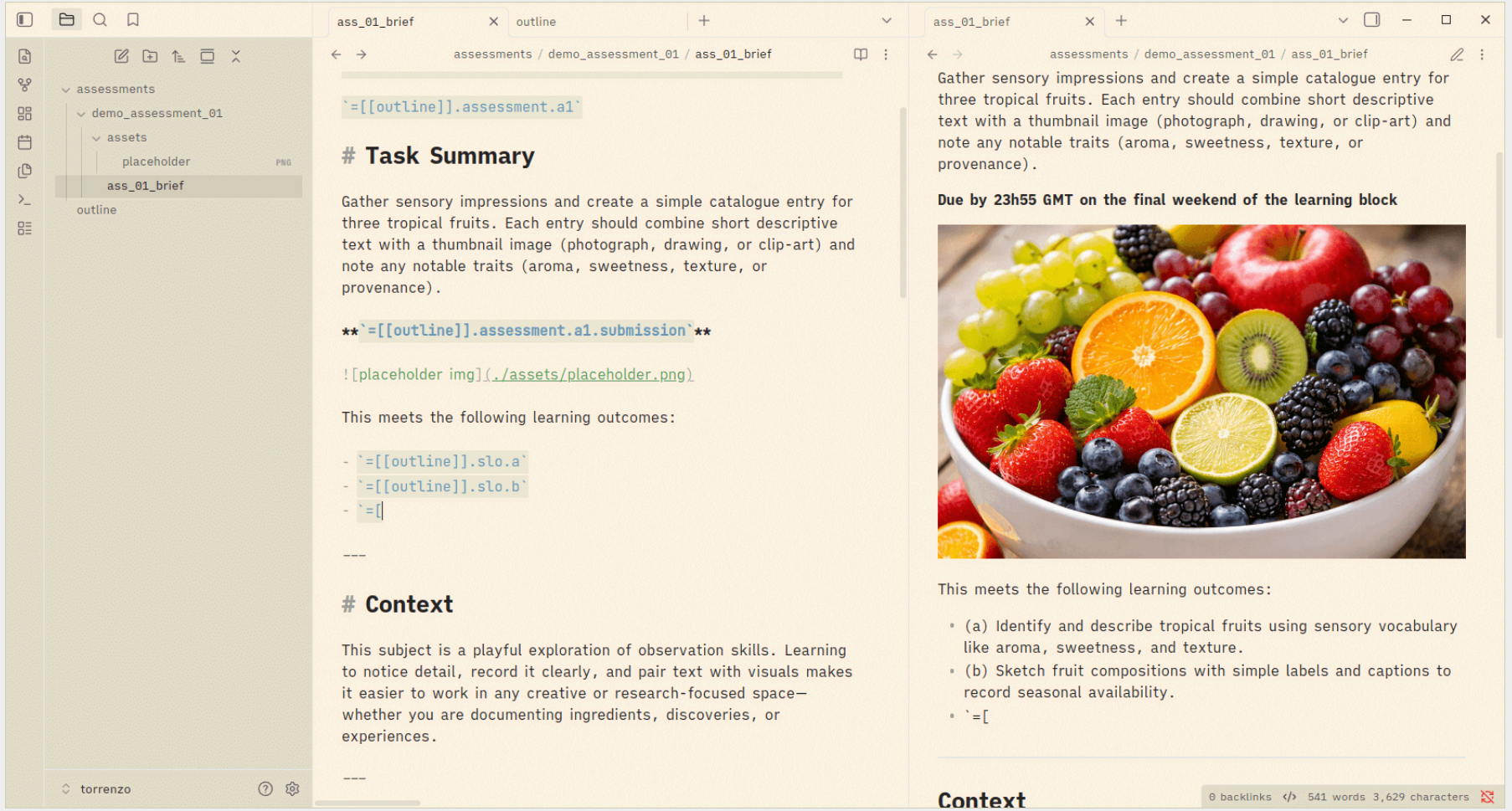
The Pipeline » Filesystem-Driven

```
your-subject/
├── outline.yaml          # single source of metadata
├── modules/              # module_<n>/mod_<n>_<seq>_<name>.md → HTML
├── assessments/          # assessment_<n>/ass_<n>_brief.md → PDF
├── includes/             # reusable snippets
├── notes/                # lecturer-only material
└── build/                # generated output
```

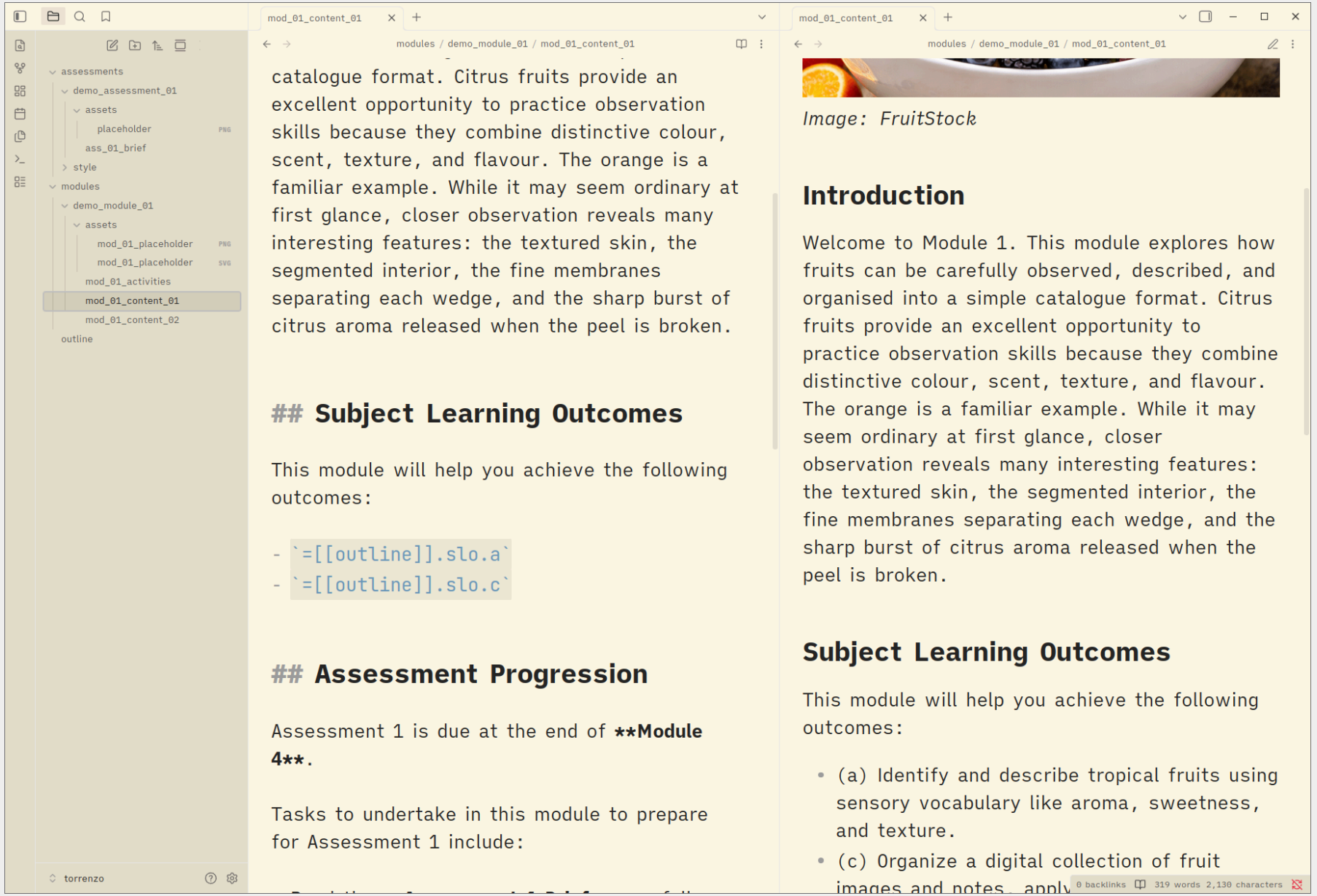
- File names and directory structure determine how content is processed

The Pipeline » Author Experience

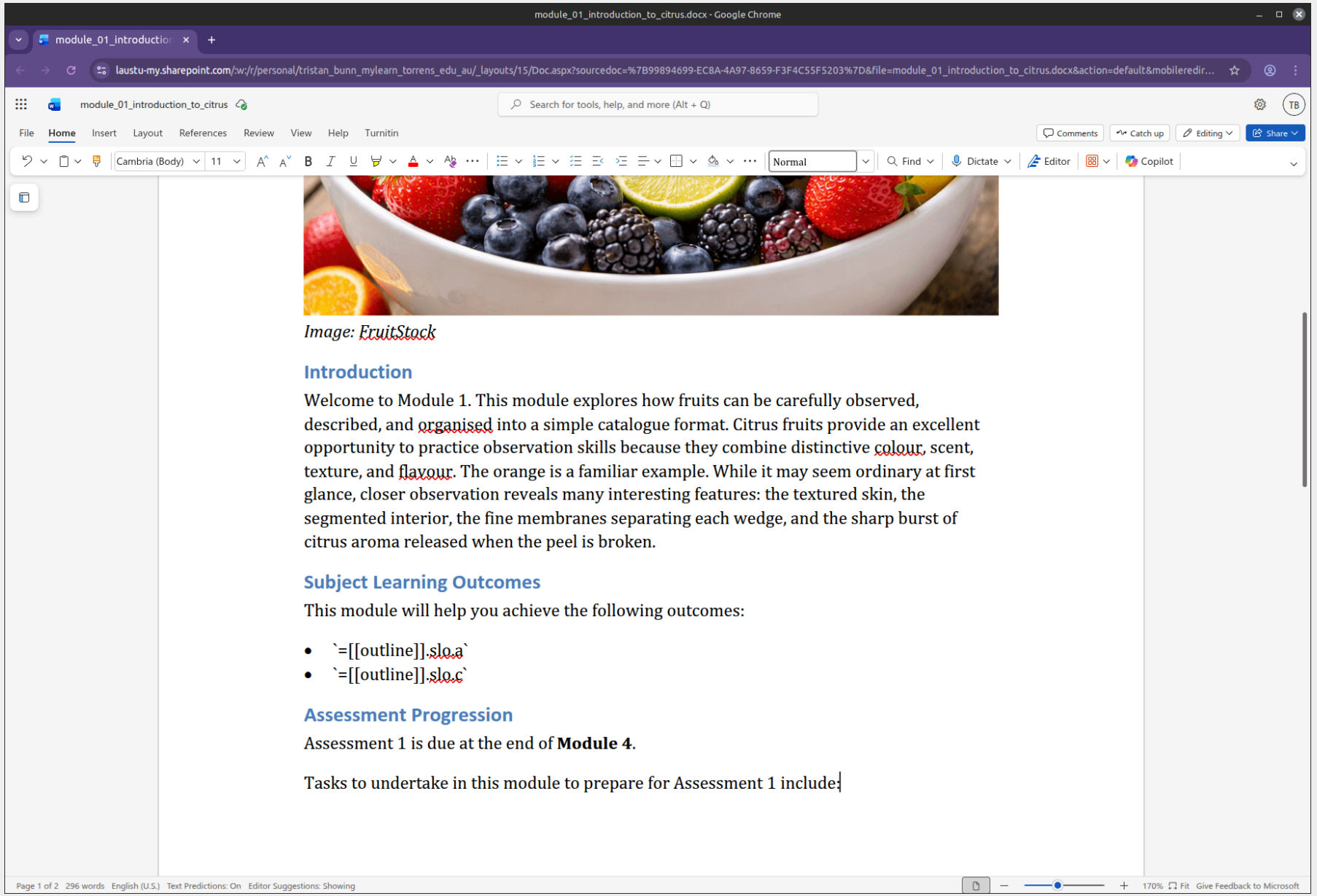
- Any editor: Obsidian, VS Code, Vim, even MS Word
- Incremental builds: only changed files rebuild; orphans pruned
- Watch mode: live-reload browser preview
- GUI: point-and-click builds (no terminal required)



editing a subject in Obsidian: wiki links, tags, plain Markdown



source mode (tags visible) vs reading view (tags resolved)



the same source, authored in MS Word

mod_01_content_03.docx - Google Chrome

mod_01_content_03.doc

laustu-my.sharepoint.com/:w:/r/personal/tristan_bunn_mylearn_torrens_edu_au/_layouts/15/Doc.aspx?sourcedoc=%7BB492728B-33D8-49A9-870B-0079EABC0EEE%7D&file=mod_01_content_03.docx&action=default&mobileredirect=true

mod_01_content_03 Review Mode

Search for tools, help, and more (Alt + Q)

TB

Back to document

Just now Save a Copy

Version History

Show edits

Just now
Current Version
Tristan Bunn

Just now
Tristan Bunn

Module 1.3: Lemons



Image: FruitStock

The Lemon

A bright yellow citrus fruit known for its intense aroma and sharp, acidic taste.

This section continues the exploration of citrus fruits by focusing on the lemon. While lemons share many structural characteristics with other members of the *Citrus* genus, they offer a noticeably different sensory experience that makes them an interesting subject for observation and description. Notably, the lemon is not a tropical fruit, so Subject Learning Outcome A is not relevant here: `=[[outline]].slo.a`

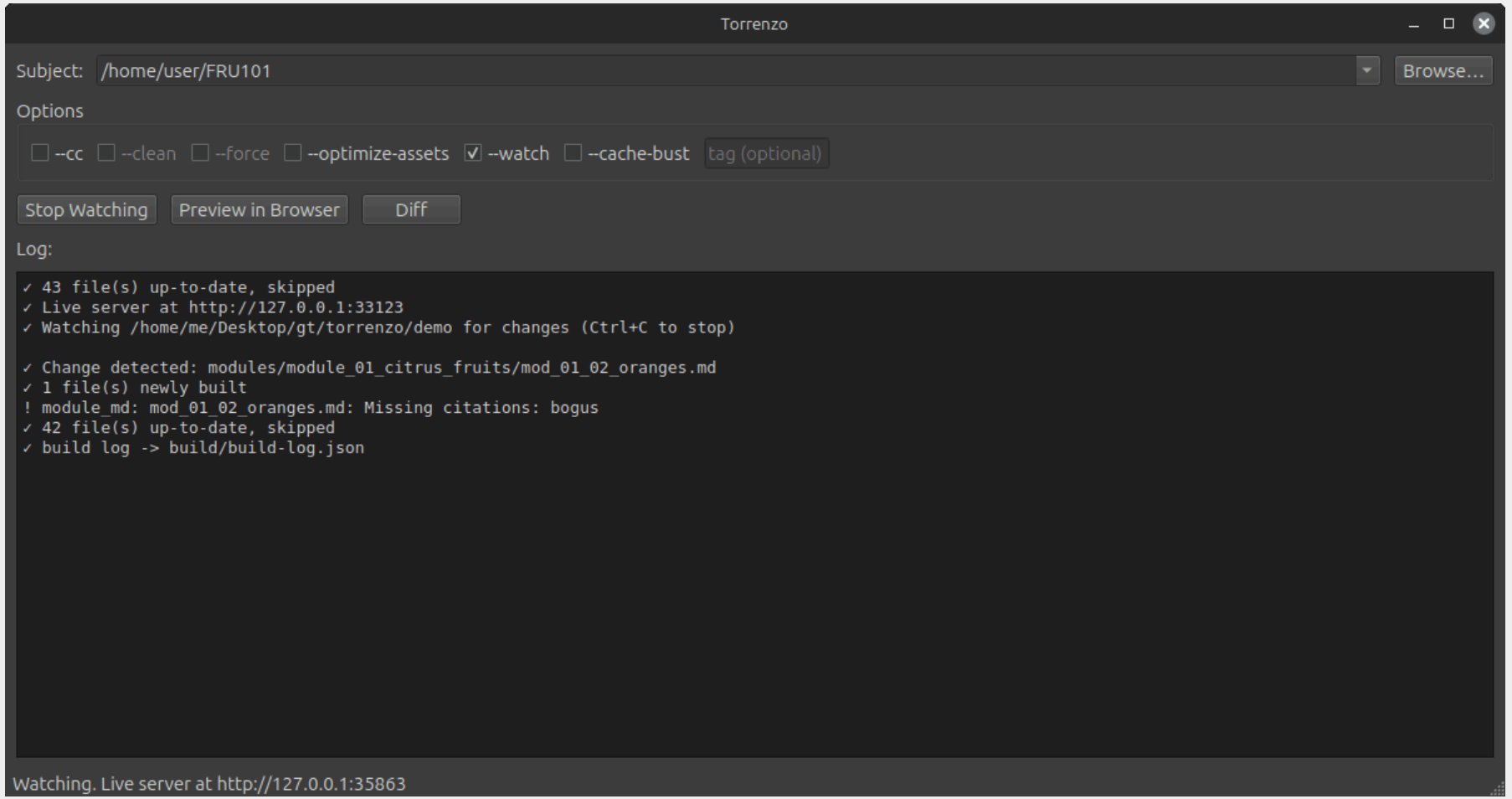
Lemons are typically smaller and more oval ~~in shape~~ than oranges, with a thick rind and a slightly pointed end. The peel contains highly concentrated aromatic oils, which produce a strong citrus scent when the skin is scratched or grated. Inside, the fruit is divided into segments filled with pale, translucent juice vesicles.

Like oranges and other citrus fruits, lemons belong to the genus Citrus.@ramirez2021citrus These fruits share common botanical features, including segmented flesh and oil-rich skins. However, lemons differ from many citrus varieties in their flavour profile. They contain relatively little natural sugar and a high level of acidity, which gives them their characteristically sour taste.

Because of this intense acidity, lemons are rarely eaten on their own. Instead, they are widely used to enhance other foods and drinks. Lemon juice adds brightness and freshness to both sweet and savoury dishes, while lemon zest contributes fragrant oils that intensify aroma and flavour.@kimura2022acidity

WordPageNumber

even Word authors get version history on tagged source



the GUI: point-and-click parity with the CLI

Patterns & Demonstrations

Patterns » Single-Sourced Metadata

- Remember the drift screenshot? **Every retyped value becomes a tag**
- Tags pull values from one outline file: subject code, outcomes, assessment details

```
This assessment is worth [[outline]].assessment.a1.weighting
of your grade for [[outline]].subject.code.
See [[mod_02_02|the module page]] before starting.
```

- Change a value **once**; every page, brief, and cartridge picks it up

Patterns » One Syntax, Five Jobs

- Outline values: `[[outline]].assessment.a1.weighting`
- Links & includes: `[[mod_02_02|the module page]]`, `[[includes|referencing.md]]`
- Module values: `[[module|n.n]]`, `[[module|name]]` -- from the filesystem
- Components: `[[component.module-navigation]]`, `[[component.video|clip.mp4]]`
- Citations: `[@harper2021sensory]` -- resolved against BibTeX

Patterns » Tags Resolve by Shape

From a real assessment brief -- even the H1 is a tag:

```
# [[outline]].assessment.a1.assessment
```

```
[[outline]].assessment.a1
```

- A **value** (`.weighting`) renders as a plain string
- An **object** (`.a1`) renders as a metadata table of all its fields
- A **list** (`[[outline]].slo`) renders as the full outcomes list

Patterns » The Filesystem Numbers the Course

`modules/module_07_data_shaping/mod_07_02_pre-class.md`

```
# Module [[module|n.n]]: [[module|name]]
```

renders as …

```
# Module 7.2: Data Shaping
```

- Numbering, titles, and navigation derive from directory and file names
- Reordering modules is a `git mv` -- every heading, cross-reference, and nav updates itself

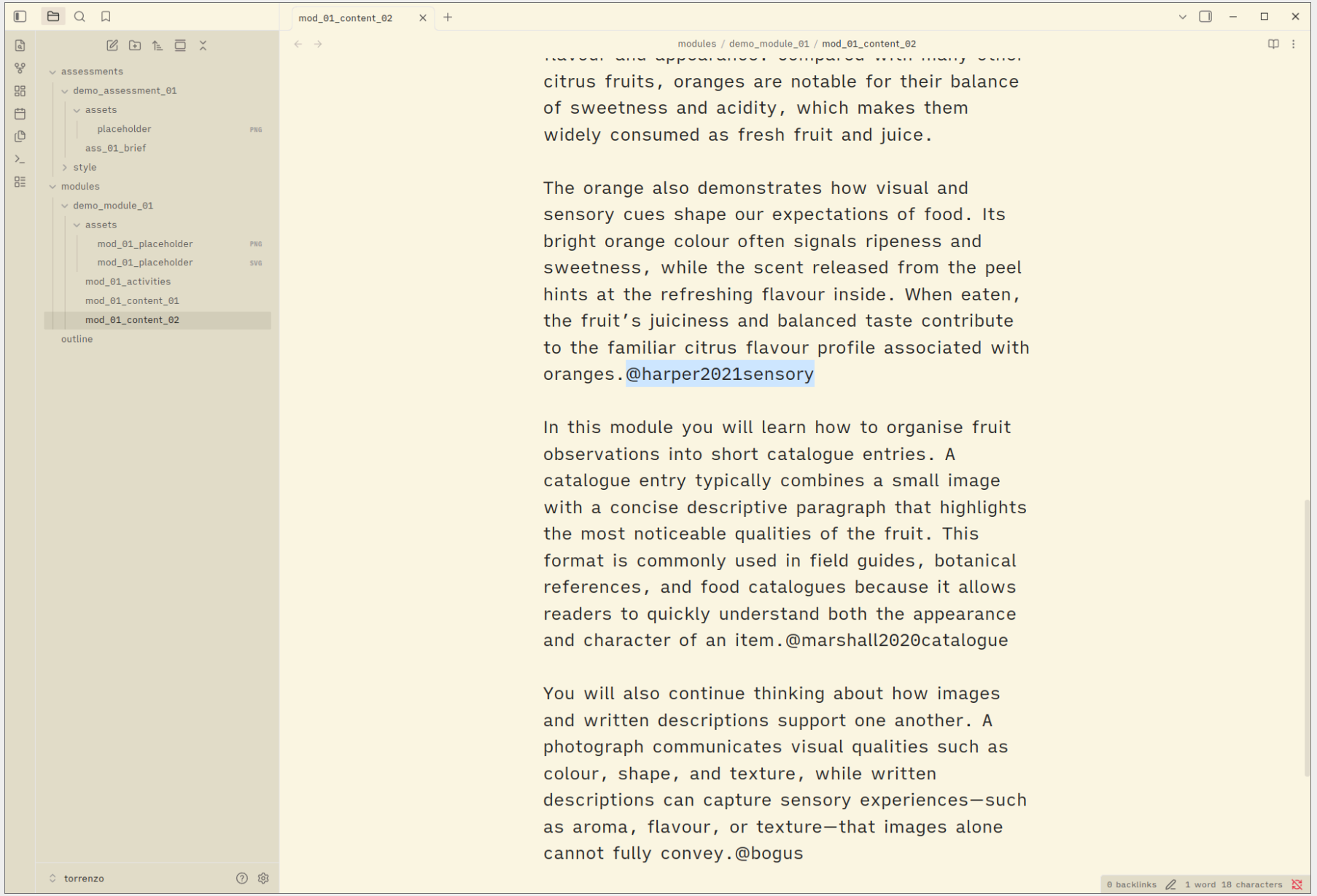
Patterns » Wiki Links & Includes

Syntax	Renders as
`[[mod_01_02_oranges]]`	*Module 1.2: Oranges* (auto label)
`[[assessment_01|Brief]]`	*Brief* (custom label)
`[[includes|referencing.md]]`	inlined reusable content

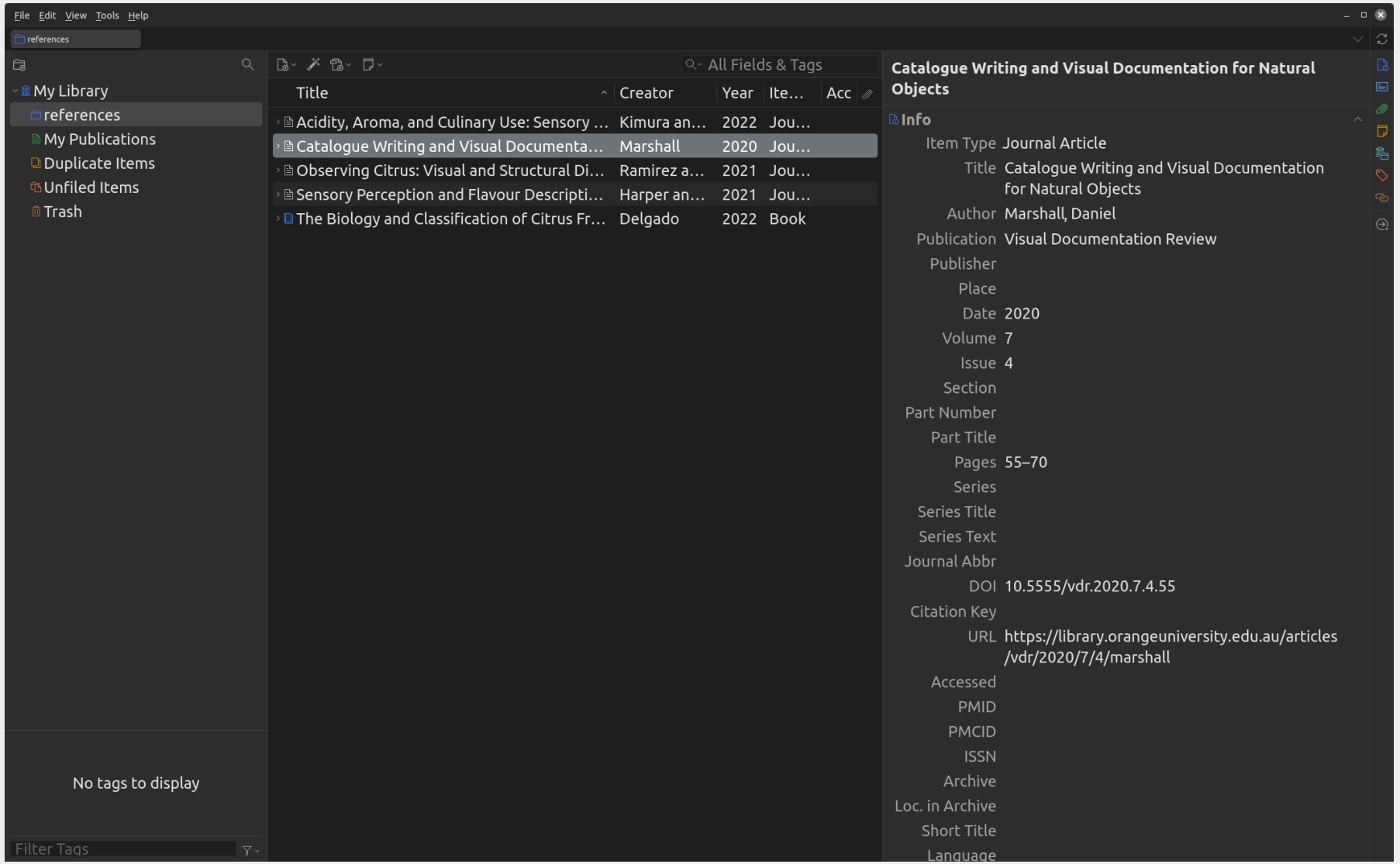
- Links rewrite per target: `.html` for browser preview, Canvas tokens for the cartridge

Patterns » Citations That Manage Themselves

- Sources live in one **BibTeX** file per subject (`references.bib`)
- In-text citations use `[@refname]`; the build renders formatted references per page
- Reference managers (Zotero, etc.) export BibTeX -- academia's existing plumbing



citation keys inline in module Markdown



a Zotero library; the pipeline consumes its BibTeX export

Patterns » Multi-Format & the Diff Tool

- Same source renders to HTML pages, PDF briefs, teaching notes, and the cartridge
- A diff tool compares a local build against a **live Canvas export**
- Answers the question: *"What actually changed on the platform?"*

Category	Meaning
* CHANGES	content, points, or checksum differ
– LIVE-ONLY	exists on Canvas, not in local build
→ REBUILT, SAME	re-rendered but identical

Around the Pipeline

Around the Pipeline » Utility Scripts

- Procedural graphics: per-module banner SVGs and course images, generated from a script -- one palette per subject
- Word sync: planner/outline `.docx` updated from `outline.yaml`, editing tracked changes in place
- PDF conversion: outline/planner documents rendered via headless LibreOffice
- Video covers: a 2-second title slide muxed onto every module video, title derived from the filename

COMPUTATION AS A MEDIUM FOR GRAPHIC OUTPUT

MONOLITHIC CMS INFRASTRUCTURE

CODE AS A LANGUAGE OF BEHAVIOR.



procedural course images: a rotated geometric pattern per subject

The Payoff

The Payoff » What Improved

- Ownership: returned to domain experts -- plain text they control
- Speed: updates to rapidly changing content, **weeks » minutes**
- Maintenance: version control, metadata, citations, and reuse all improved
- AI-assisted review: enabled by structured, local source material

There is room for more or less automation -- the pipeline can be calibrated per subject and team.

The Payoff » A Complete, Local Project

- **Search everything** -- file manager, editor, or `grep` across all source
- **Automated and batch checks** -- broken links, missing citations, orphans
- **AI review** -- flag outdated content, references, or material needing clarity

orange

Aa

Collapse results

Show more context

Explain search terms

15 results

File name (A to Z)

mod_01_activities 5

Access [Observing Everyday Objects](https://orangeuni.edu/library)

...r provides a short overview of citrus fruits and their distinctive characteristics. It explains how oranges, lemons, mandarins, and limes differ in flavour, acidity, aroma, and visual appearance. The chapte...

Access [Citrus Diversity and Characteristics](https://orangeuni.edu/library)

Access [Writing Catalogue Entries](https://orangeuni.edu/library)

Access [Photographing Food and Natural Objects](https://orangeuni.edu/library)

mod_01_content_01 1

...actice observation skills because they combine distinctive colour, scent, texture, and flavour. The orange is a familiar example. While it may seem ordinary at first glance, closer observation reveals many ...

mod_01_content_02 9

Module 1.2: Oranges

The Orange

...ce because they combine vivid colour, distinctive scent, varied textures, and complex flavours. The orange provides an excellent starting point.

Although oranges are familiar to most people, careful observation reveals many details that often go unnoticed. The...

Oranges belong to the genus *Citrus*, which also includes lemons, limes, mandarins, and grapefruits.@delga...

...ns—but they differ significantly in flavour and appearance. Compared with many other citrus fruits, oranges are notable for their balance of sweetness and acidity, which makes them widely consumed as fresh ...

The orange also demonstrates how visual and sensory cues shape our expectations of food. Its bright orange col...

...it's juiciness and balanced taste contribute to the familiar citrus flavour profile associated with oranges.@harper2021sensory

torrenzo

mod_01_content_01

modules / demo_module_01 / mod_01_content_01

mod_01_content_01

Module 1.1: Introduction to Citrus Module



Image: FruitStock

Introduction

Welcome to Module 1. This module explores how fruits can be carefully observed, described, and organised into a simple catalogue format. Citrus fruits provide an excellent opportunity to practice observation skills because they combine distinctive colour, scent, texture, and flavour. The orange is a familiar example. While it may seem ordinary at first glance, closer observation reveals many interesting features: the textured skin, the segmented interior, the fine membranes separating each wedge, and the sharp burst of citrus aroma released when the peel is broken.

Subject Learning Outcomes

This module will help you achieve the following outcomes:

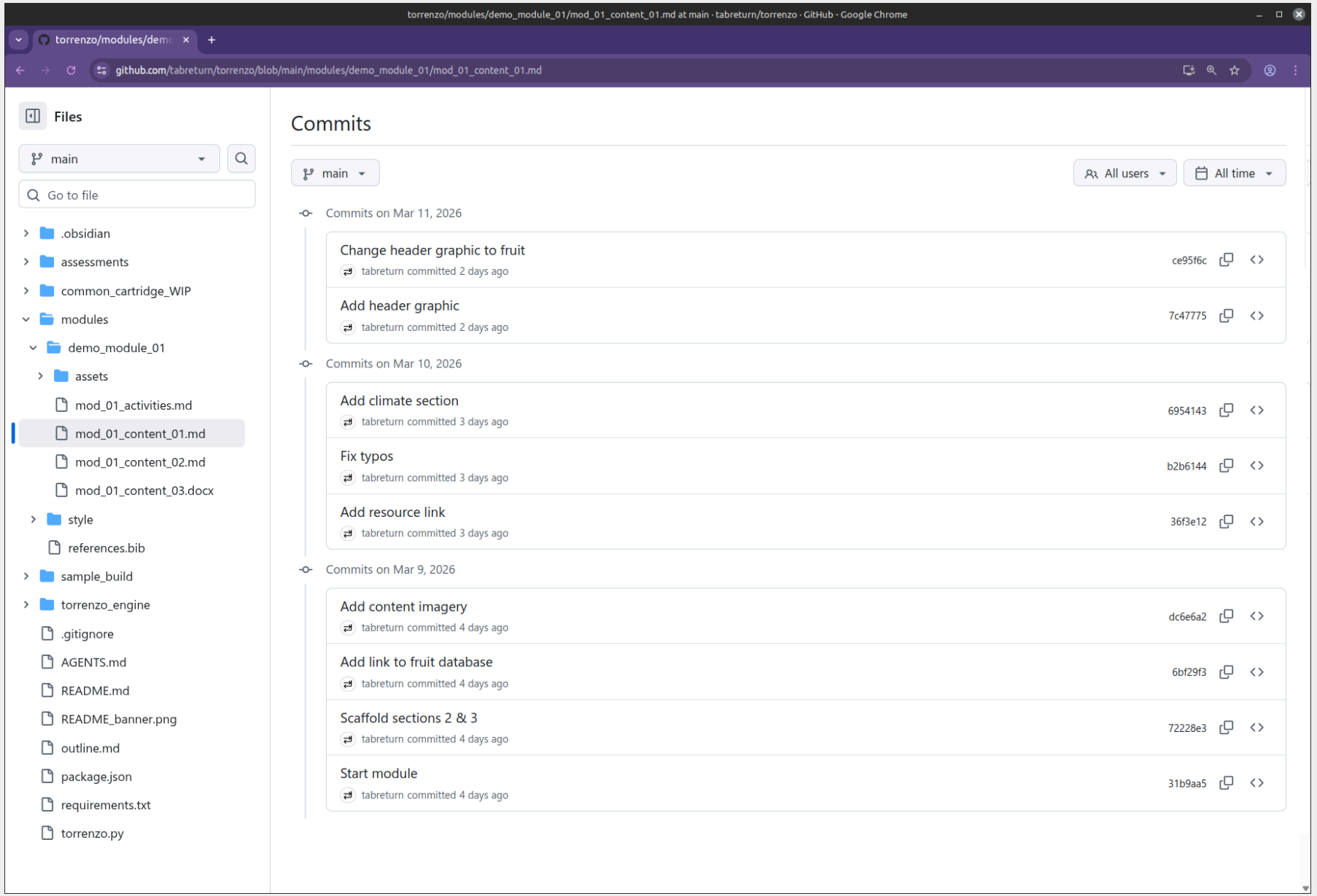
- (a) Identify and describe tropical fruits using sensory vocabulary like aroma, sweetness, and texture.
- (c) Organize a digital collection of fruit images and notes, applying a consistent tagging scheme for easy reference.

0 backlinks 319 words 2,130 characters

one search across every module, activity, and brief



an AI coding agent working directly on course source



commit-level answers to "what changed, when, and why"

What Broke

What Broke » Quirks & Limitations

- Canvas import: unexpected overwrite behaviour (not additive), duplicated assets after re-import
- Image caching: images sometimes stop rendering after re-import; worked around with cache-busting asset suffixes
- Word support: weak on the authoring side
- Final 5 percent: manual cleanup after each import
- Compliance: friction accommodating the university process

Takeaways

Takeaways

- Treat the LMS as a deployment target, not an authoring environment
- The repository is the source of truth; builds are reproducible
- Plain text + a build pipeline returns ownership, speed, and reviewability
- Expect platform quirks -- budget for a manual final 5 percent

End